AML FOOTBALL DOC’S AND USER’S MANUAL

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PLATFORM IDEOLOGY

AML Football sits at the intersection of football, machine learning, and mathematics. More than a goal prediction platform, AML Football is built on a single belief: that AI and mathematics can meaningfully change how we understand and analyse the beautiful game.

It started as a weekend project, inspired by *Moneyball*, a film about using mathematics and statistics to find hidden edges and advantages in sport. That sparked a question: could data science do the same for football? Not for gambling, but to break down what actually makes teams win, what gives them momentum, and whether the hidden patterns inside the sport we love could be unlocked with data and AI.

The problem every football fan or analyst faces is simple: we want to know what to expect from a match before it happens. Not just who might win, but the shape of the game itself, the scenarios, the possibilities, the probabilities. AML Football is built to solve that.

We are starting with goal related predictions, but the vision goes further, position predictions, player heatmap forecasting, and every meaningful way AI can be applied to football. This is version one of something much bigger.

Who It's For:

AML Football is built for football fans, analysts, data professionals, and ML/AI enthusiasts who believe the game deserves more than gut feeling or guessing and want to see what happens when AI meets football.

Test/Evaluation Results

We carried out extensive testing to evaluate how reliable the AML Football system is.

Initially, the goal was to predict match outcomes i.e win, lose, or draw. After multiple training runs and modelling approaches, every attempt fell short of a reliable accuracy threshold. Predicting match outcomes in football is notoriously difficult even for the best models in the world, and we were not going to ship something we couldn't trust.

Below are results based on test data that have not been seen by the model:

1. **Initial Accuracy for Three Way Outcome:**

   Evaluation was carried out on both Laliga and Serie A models possible 3 way outcome accuracy.

   Laliga model recorded a 42% accuracy and Serie A recorded a 47% accuracy which is slightly higher but still generally low for real world performance. This metrics tells us how accurate the model can be in real world scenario, with this current performance if a team is predicted to win it is only 42 or 47 percent sure of that prediction for Laliga and Serie a respectively.

```
5]: accuracy = accuracy_score(laliga["Actual_Outcome"], laliga["Predicted_Outcome"])
    accuracy

5]: 0.42395437262357416
```

(Laliga Accuracy Score)

```
9]: accuracy = accuracy_score(seriea["Actual_Outcome"], seriea["Predicted_Outcome"])
    accuracy

9]: 0.4742857142857143
```

(Serie A Accuracy Score)

2. Regression Metrics:

To evaluate how accurately the model predicts goals, we used two standard regression metrics:

Root Mean Squared Error (RMSE): Calculated by subtracting the predicted goals from the actual goals, squaring the result, then taking the square root to make the figures absolute. RMSE penalises large errors more heavily, meaning the model is punished significantly for predictions that are wildly off.

Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual goals. This gives a more intuitive view of how wrong the model typically is on any given prediction, think of it as the average margin of error per match.

Below shows Laliga and Serie A has a almost a similar result, it is basicly saying if the model predicts 2 goals for a team on a normal day, if the model is to be wrong the actual goal might be 3 or 1 goal.

```
Model Evaluation Metrics (per match):
-------------------------------------
Home Goals -> MAE: 0.96, RMSE: 1.20
Away Goals -> MAE: 0.99, RMSE: 1.33

Notes:
- MAE shows the typical error per match (average goals off).
- RMSE shows typical error magnitude, giving more weight to larger mistakes or model prediction error.
```

(Laliga Metrics)

```
Model Evaluation Metrics (per match):
-------------------------------------
Home Goals -> MAE: 0.86, RMSE: 1.07
Away Goals -> MAE: 0.88, RMSE: 1.11

Notes:
- MAE shows the typical error per match (average goals off).
- RMSE shows typical error magnitude, giving more weight to larger mistakes or model prediction error.
```

(Serie A Metrics)

3. Acceptable Prediction Rate:

I didn't expect us to achieve perfect accuracy because football is filled with unpredictable events that can decide how a game eventually ends. I planned to manage the errors and be able to use them to our advantage instead.

To measure how accurate the system can be, we focused on evaluating how consistent the model was in predicting actual goals, +1 goals and -1 goals.

This evaluation showed Laliga having an average of approximately 80% and Serie A having approximately 87% accuracy.

```
[61]: count = laliga["Home_Error"].isin([0, 1, -1]).sum()
      print(count/len(laliga))

      0.7965779467680608
```

(Laliga 0,+1 & -1 Error Accuracy)

```
[15]: count = seriea["Home_Error"].isin([0, 1, -1]).sum()
      print(count/len(seriea))

      0.8647619047619047
```

(Serie A 0,+1 & -1 Error Accuracy)

CONCLUSION:

We focused on this result as our evaluation metrics because it directly supports the statistical and mathematical way our system was built.

USER'S MANUAL & GUIDE

The following section is a complete guide to using the AML Football platform. It explains every number, label, and recommendation shown in the app so that both technical and non-technical users can get the most from the data.

1. What Is AML Football?

AML Football is an AI platform that applies machine learning and statistical analysis to football fixtures. It examines the recent form of both teams in a match and produces a set of data-driven findings about what to expect, with a focus on goals. It is not a tips or gambling service. It is a research and education tool built for football fans, data enthusiasts, and anyone curious about what AI can reveal when applied to the beautiful game.

2. The Main Feed (Dashboard)

When you open the app, you land on the main feed. It shows all the active fixtures, grouped by league. Each match appears as a row. Here is what you will see on each match row:

Team Names: The home team is listed on top and the away team below it, each with their initials shown on the left.

Primary Recommendation: Below each match row, the app shows the single most statistically supported finding for that fixture, written in plain English. The key part is always shown in bold and includes a Sureness Score (for example: 'Away side goals 1-2 (99% sure)'). This is the app's strongest data signal for that match.

View Analysis button: Clicking anywhere on the match row opens the full analysis page for that fixture.

Note: The main feed deliberately shows no scoreline or goal predictions. The only output here is the Primary Recommendation, which is the most reliable and honest single finding the system has found for that match.

3. The Primary Recommendation & Sureness Score

The Primary Recommendation is the single most statistically consistent finding for a fixture. It is selected purely by which outcome the data backs most strongly, based on both teams' recent history. It always appears with a Sureness Score.

The Sureness Score is a percentage that tells you how consistently this particular pattern has appeared across both teams' last 10 games. Here is how to read it:

90% - 99%: Very strong pattern. This outcome has been extremely consistent in recent matches.

70% - 89%: Solid finding. A reliable trend, but allow for some variability.

50% - 69%: Moderate finding. The data leans this way, but it is not a standout signal.

Below 50%: Weak signal. More uncertainty than confidence. Treat with caution.

4. The Full Analysis Page

Clicking a match row opens the Analysis Page, which organises all findings into three tabs: Match Summary, Full AI Forecast, and Team Insights.

4a. Match Summary Tab

This is the recommended starting point for any match. It contains four key sections:

ML Auditor Report: A short paragraph written in plain English by the AI. It explains what pattern stood out most and why it was identified as the primary finding for this match.

Forecast Confidence: The single highest-confidence Over/Under finding for the match, shown as a card with its Sureness Score. This is the O/U signal the data backs most strongly.

Expected Total Goals: The most probable total goal range for the entire match, shown as a single card. For example, '2-3 Goals at 65% Confidence' means the data most consistently points to 2 or 3 goals combined between both sides.

Team Performance Forecast: Two cards side by side - one for the home team and one for the away team. Each shows the most probable goal range for that team individually and its confidence level.

4b. Full AI Forecast Tab

This tab provides the complete breakdown of every statistical scenario the system has generated.

ML Market Confidence: Shows all Over/Under probabilities - Over 1.5, Over 2.5, and Under 3.5 - each with their individual confidence percentages.

All Total Match Scenarios: Three levels of total goal scenarios for the match, expanding outward in range. ML Simulation Level 1 is the tightest and most precise. Level 2 is a slightly wider range. System Reference (Level 3) is the 95% historical window showing the broadest expected range.

Detailed Team Projections: Shows all goal range scenarios for the home team and away team separately, each at multiple levels of precision.

4c. Team Insights Tab

A focused breakdown of each team's individual goal range predictions at every scenario level. Use this tab to compare how predictable each team is individually. A team showing a very tight range (such as '1-1') at high confidence is highly consistent. A team with a wide range across all scenarios has been more volatile in recent games.

5. Understanding Goal Intervals

Goal intervals are ranges of goals rather than single exact numbers. They appear throughout the app and are the core of what AML Football produces at it beta stage. Here is how to read them:

A tight interval like '1-1' or '2-2': The data is very concentrated. The team has scored the same number of goals consistently. A high confidence on a tight range is the strongest kind of finding in the app.

A wider interval like '1-3': The data covers more outcomes. The team's scoring has varied more. The probability is naturally higher because more results qualify, but it is less precise but naturally probable as a finding.

Home team goal interval: The most probable range of goals for the home side only, based on their recent matches.

Away team goal interval: The most probable range of goals for the away side only.

Total goals interval: Combines both teams. This is the expected range for the match as a whole and is generally the most reliable figure to focus on.

Tip: Always read the interval alongside its confidence percentage. A tight interval at 80% is a far stronger signal than a wide interval at 80%.

6. Over / Under Explained

The Over/Under figures show the probability of the match crossing common goal total thresholds. You will see three of these in the app:

Over 1.5 Goals: The probability that the match produces 2 or more goals in total. Results like 1-1, 2-0, or 0-3 all qualify. This is almost always the highest figure.

Over 2.5 Goals: The probability of 3 or more total goals. Results like 2-1 or 3-0 count. A more selective threshold - a high score here suggests an open, high-scoring game is likely.

Under 3.5 Goals: The probability of 3 goals or fewer in total. A high Under 3.5 alongside a high Over 1.5 describes a moderate-scoring match - goals expected, but not a high-scoring affair.

Example: 'Over 1.5 at 96%, Under 3.5 at 84%' together say this match will almost certainly have goals, but is unlikely to be a goal fest. The sweet spot is 2-3 goals total.

7. Confidence Colour Codes

Every probability figure in the app is colour-coded to help you quickly judge how reliable a finding is:

Green (90% and above): Very high confidence. The data is extremely consistent for this outcome.

Amber (50% - 89%): Moderate confidence. A clear trend exists but allow for some variability.

Red (below 50%): Low confidence. The data is mixed or the outcome is unlikely.

8. How to Get the Most From AML Football

Start with the Primary Recommendation on the feed. This is always the data's strongest single finding. If you want one clear takeaway from a fixture, this is it.

Check the Sureness Score. Anything above 85% is a strong signal. Below 60%, treat it as a guide rather than a certainty.

Open the Match Summary tab first. The ML Auditor Report and the Expected Total Goals card give you the clearest, most reliable picture of the match in the fewest steps.

Cross-reference the Over/Under with the Total Goals interval. If both point in the same direction, that convergence is a significantly stronger signal than either one alone.

Use the Full AI Forecast and Team Insights tabs for deeper analysis. These tabs are for users who want to see every scenario the system has generated, not just the headline finding.

APP LIMITATIONS AND RISKS

After careful analysis and monitoring, we identified that the model's reliability drops significantly when unexpected in-game events occur such as an early red card, unusual weather conditions, or an unusual goal outburst, particularly from the home team. These scenarios fall outside the patterns the model was trained to recognise and can significantly alter the natural flow of a match.

This remains one of our key challenges, and we are actively working to incorporate relevant data to better account for these situations in future iterations.

CONCLUSION

AML Football was built on a simple but ambitious idea: that mathematics and machine learning can give football fans and analysts something they have never had before, What they should expect before a match kicks off.

Version one proves the concept. With an Acceptable Prediction Rate of approximately 80% for La Liga and 87% for Serie A. It is treated as a system built on probability, trained on data, but like any system, it has it limitations.

For this first version, performance will be judged primarily on accuracy of the LLM’s recommendations. This gives the system the best conditions to demonstrate its true reliability before we expand evaluation across all tiers in future iterations.

This is just the beginning. Goal prediction is the foundation, but the vision for AML Football extends into position predictions, player heatmap forecasting, and every meaningful intersection between AI and the beautiful game.

The unpredictability of football is not a problem to be solved, it is what makes the sport worth watching. AML Football does not claim to remove that uncertainty. It simply helps you understand it better.